

SECTION-VI
REMOTE SENSING

Remote Sensed Data Analysis for Geology and Glacial Geology of Wohlthat Muhlig-Hofmann Mountain Chain in Queen Maud Land, East Antarctica

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Abstract

During the Fifth Indian Scientific Expedition to Antarctica, aerial surveys were conducted over Schirmacher hills; Gruber and Petermann massives in eastern part of Wohlthat mountain range in Queen Maud Land. Spectral signatures of sea-ice, fast-ice, shelf-ice, continental-ice and lithological units were collected. Maps on 1:2,50,000 scale for the region covering the area from 3°E to 21°E, encompassing the shelf-ice coast in the north and Muhlig-Hofmann mountains, Conrad mountains and the Wohlthat mountains towards the south have been prepared. The main drainage glaciers, the local glaciers and their associated morainic deposits have been shown. Sastrugi fields, nunataks, solid crystalline blue ice fields, their distribution, melt water channels, and grounded ice have been demarcated. Thermal mapping of Schirmacher hills has also been conducted.

Digital analysis to classify the rock types, to enhance the structural features, to differentiate the occurrence of various types of ice sheets and locate the solid crystalline ice fields which are typically associated with nunataks and mountains, has been carried out.

Introduction

Satellite imagery provides a synoptic and repetitive overview and multiband digital data enables extraction of information about surface features and types of cover. With the advent of remote sensing techniques it has become quite possible to map remote regions with minimum of ground visits for terrain evaluation and natural resources inventory. It is very useful in Antarctic studies where conventional exploration is pursued at great expense and risk.

Interpretation of the LANDSAT Multi Spectral Scanner imagery for morphology of shelf-ice, continental ice-sheet, glacial flow directions, fracture systems, and boundary demarcations of different types of ice-sheets has been carried out. During the Fifth Indian Scientific Expedition to Antarctica aerial surveys have been conducted over Schirmacher hills, Gruber and Petermann massives in eastern part of Wohlthat mountain range in Queen Maud Land of Antarctica. Spectral

signatures of sea-ice, fast-ice, shelf-ice, continental-ice and lithological units have been collected.

Maps on 1:250,000 scale for the region covering from 3 degrees East to 21 degrees East longitude encompassing the shelf-ice, Muhlig-Hofmann, Conrad and the eastern Wohlthat mountain range have been prepared. The main drainage glaciers, the local glacier distribution and associated morainal deposits have been depicted. Sastrugi fields, nunataks, solid crystalline blue ice fields, their distribution, melt water channels and grounded ice bergs have also been demarcated.

Digital analysis has been carried out to classify rock types, enhance the structural features, differentiate the occurrence of various types of ice sheets and locate the solid crystalline ice fields typically associated with nunataks and mountains, signifying negative ice budget regimes. Digital analysis for rock type classification has been done and maps on 1:50,000 scale for Wohlthat mountains and on 1:25,000 scale for Schirmacher hills have been prepared. The coastal i.e., shelf-ice configuration, tensional cracks along the edges of shelf-ice have been mapped.

Thermal mapping of Schirmacher hills has been conducted. The Precision Radiation Thermometer temperatures recorded over the landmass varied between 2 degrees Celcius and 21 degrees Celcius.

. Relatively smooth surface south of mountain range without the crevasses indicates that the ice thickness in these regions is less. This region is interpreted as having shallow basement which is gently sloping towards south. The northern face of the chain is suspected to have a deep seated fault, giving rise to greater thickness of the ice.

Ground/Ice truth data has been collected over an area covering one Landsat scene of 185x185 km, which was incorporated in interpreting the imageries and carrying out digital analysis to derive maximum information of the environs around Indian Antarctic Scientific Research Station, DAKSHIN GANGOTRI.

Objectives of Remote Sensing Experiments

The objectives were to analyse remote sensed data to demarcate various morphological units, decipher ice types and boundaries, locate zones of crevasses, fractures, classify the rock types, collect ground/aerial remote sensing data in clear atmospheric conditions and to extract all the retrievable information from the satellite data and prepare maps depicting glacial morphology and geological classification.

Equipment:

1. Exotech 100-A radiometer: Two units

The four visible wavelength windows which are similar to those of Landsat Multi Spectral Scanner have been used. The bandwidths are as follows:

MSS

BAND 4 0.5 to 0.6 Micro metres

BAND 5 0.6 to 0.7 Micro metres

BAND 6 0.7 to 0.8 Micro metres (visible near infrared)

BAND 7 0.8 to 1.1 Micro metres (visible near infrared)

2. Precision Radiation Thermometer

This was utilised to measure sea surface temperatures on various lithological units. Ground level observations were taken at Schirmacher and air borne recording over Wohlthat mountain range and Schirmacher hills.

Ground Truth Data Collection

Ground based surveys and observations were carried out on shelf-ice and at Schirmacher hills while traversing the rock formations. The temperatures were measured on rocks which varied from 2 degrees Celcius to 21 degrees Celcius for leucocratic to melanocratic rocks and variations related to mineral assemblages.

Aerial surveys with Radiometer and Radiation Thermometer were conducted on Schirmacher hills and eastern Wohlthat range (Table I). Further 980 line km survey was conducted over shelf-ice, the grounded Kaliningrad Ice, sastrugi field, ablated and crystalline, polar/continental ice, melt water bodies, cirque regions, refrozen ice, nunataks, flow structures, crevassed regions, morainal grounds and fast ice along the shelf-ice coast.

Table I

Region Surveyed	Line Km Flown	Area Covered in Sq Km
Schirmacher hills and adjacent ice-field	250	600
Gruber and Petermann massives	650	1,200
Reconnaissance survey from Dakshin Gangotri to East, SW, Schirmacher hills, NW, and Profiles, including intermittent flights over blue ice fields, and Nunataks	980	—
Total	1,880	1,800

Methodology**Visual Interpretation**

Landsat MSS multiband image interpretation has been carried out to discriminate glacial morphological units, ice types, crevasses, morains deposited and

glacial flows. Tone, texture and shape characteristics offer limited scope in the analysis of the black and white imagery of Antarctica.

Because of poor contrast due to the predominance of ice/snow cover the exposure times had to be considerably increased while making the prints to bring out the subtle features. An attempt was made to produce false colour composite using bands 4, 5, and 7 and the result was very poor, because the entire grey levels are concentrated at the brighter end of the grey scale and only usage of digital enhancement techniques produced good false colour images, (Figs. 1 and 2). Visual interpretation of black and white imagery and glaciomor-



Fig 1

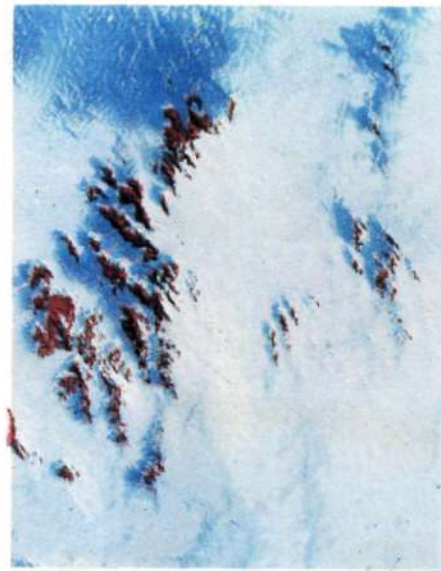


Fig. 2

phological studies were carried out on 1:250,000 scale (Fig. 3). The following features have been interpreted and delineated:

Solid crystalline blue ice fields, fracture systems, glacial flow lines, drift ice dunes, crevasses, wind scoops, ice boundaries, topographic highs, drift ice cover over blue ice fields, exposed isolated peaks of hills (Nunataks) or mountains and shallow basement relief. The sun elevation angles ranged between 22 degrees and 29 degrees. Elevation differences of the grounded ice berg regions and crevasses have been brought but clearly due to low elevation angles. The tonal variation is very limited and the darker tones appearing in various patterns signify the absorption levels of various ice types.

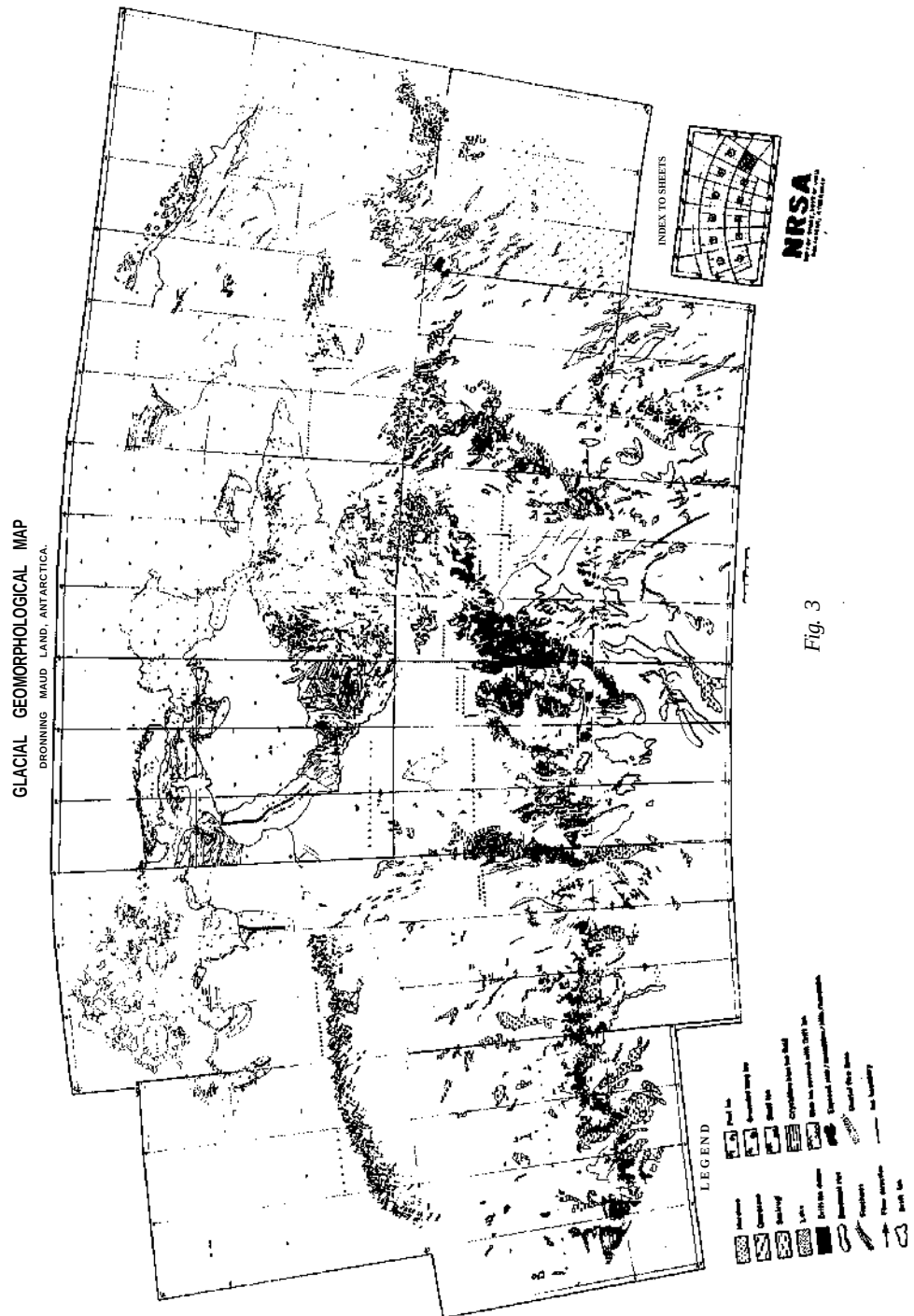


Fig. 3

The fractures at shelf-ice edge were in contrast to the polar and shelf-ice relief and could be marked accurately. The localised elevations representing the grounded ice were identified as the sun lit side of the hump was in contrast to the shadowed region and the ridge of the ice could be marked. The shelf ice edge is discernible from sea-ice as the former is elevated by 10 to 30 metres and shows up as a contrasting edge. The boundaries of shelf ice and the continental ice sheet have been delineated from the smooth surface and rugged surface by textural and tonal variations.

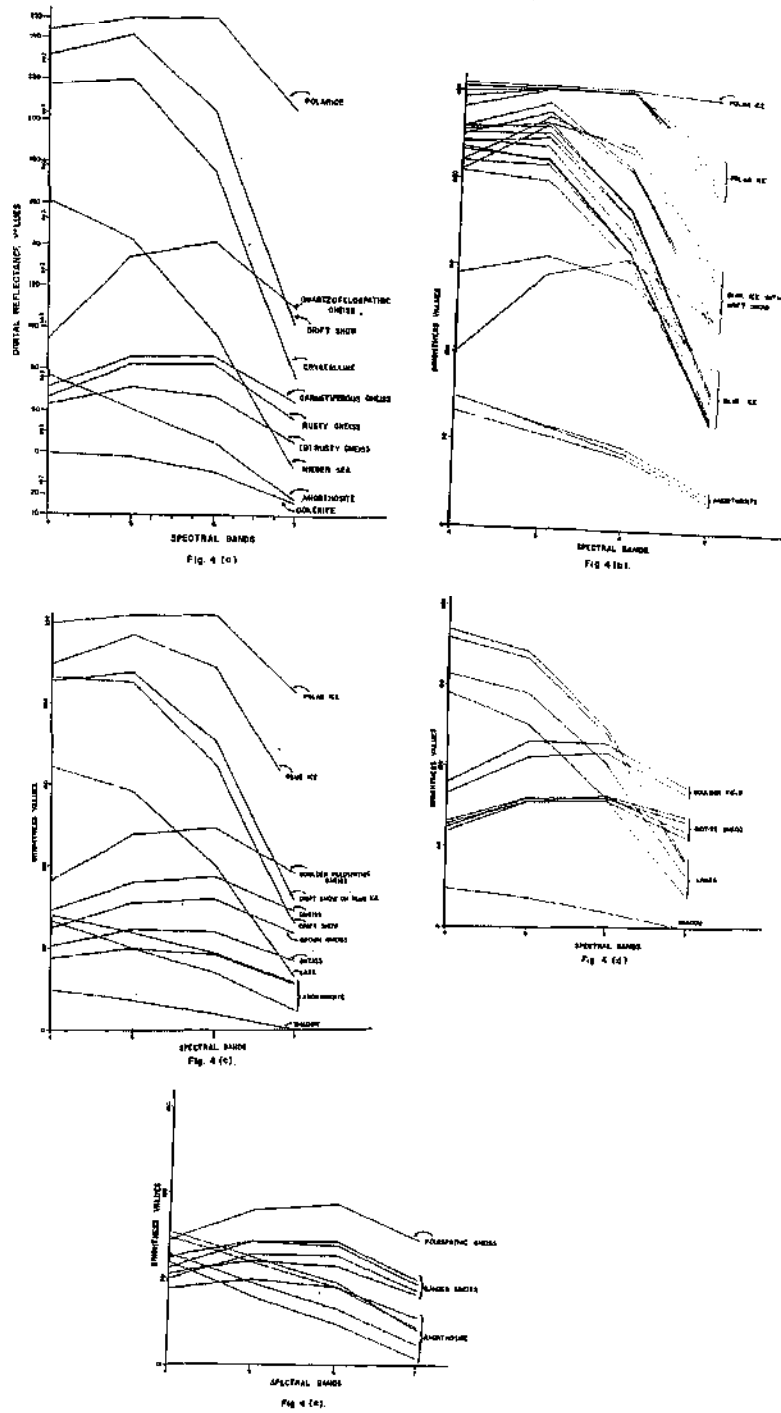
The shelf ice possesses relatively uniform tone and texture because it is devoid of large scale crevassing. The crevassing pattern on the shelf-ice edges is continuous and linear. The crevassing pattern of continental ice sheet is irregular and marked by low frequency repetitivity. The reflectance properties of the shelf-ice and continental ice vary due to the surface roughness. The dark regions on the continental ice are typical regions where the top cover of loose drift ice is removed exposing the underlying solid crystalline blue ice. The bright tones followed by the dark tones indicate differential ablation and the drift snow dunes in arcuate shapes occur with regularity.

Computer Aided Analysis

Supervised classification has been carried out using nearest neighbourhood algorithm. Training sets on Gruber massif, based on field data, were fed to Multispectral Data Analysis System for collecting statistical parameters for each class. Ratioing and digital enhancement techniques have been used to bring out every detail on the imagery. The feldspathic gneisses, boulder fields, banded gneisses, granitoids, anorthosites, lakes, shadows, polar-ice, crystalline blue ice and drift ice training sets were fed. All the signatures have been compiled and presented in Fig. 4 a, b, c, d, e. Though the purity of the training sets finally fed ranged from 70% to 100% the classification achieved is optimal under the limitations where the snow cover interference and shadow influence contribute for misclassification.

Analysis

The landsat MSS band 7 data has been interpreted visually to extract the morphological features by comparing the band responses. Though the entire image consists of the ice field, the tonal variations bring out the subtle features and the differential absorption of light in the four bands brings out the large scale crystalline ice fields. The following major physiomorphic zones have been delineated. 1. Coastal zone: (a) Sea-ice pack with one to two metres thick ice or continental glacier (b) Shelf-ice cliff with 10 to 40 metres exposed above the mean sea level. (c) Continental glacier directly flowing into the sea. 2. Zone of glacial sheet grounding or grounded ice bergs in shallow waters where ridges are present. 3. Zone of shelf-ice bounded by the continental glaciers. 4. Zone of continental glaciers along shelf. 5. Zone between high slopes of (a) Continental



ice-sheet and rising mountain region, (b) Intraglacial plateau region. 6. High mountainous regions 7. Inter-mountain glaciers with steep slopes. 8. Highly elevated glacier ice sheet plateau gently rising towards the inner (Southern) continental part. 9. Buried mountain massives with ridges or plateau relief. 10. Isolated peaks of hills and mountains (Nunataks). 11. Shallow basement regions.

Ice Types and Band Responses

Polar Ice Sheet

In all the four bands polar ice is characterised by very high reflectivity. The drift ice forms the top layer consisting of fine crystals/granules of ice. Multiple reflections taking place between the fine grained particles cause Rayleigh scattering, and enhance the radiance levels to such an extent that the sensor saturation levels are reached (O'Brien and Muris, 1975). The continental ice sheet consists of numerous crevasses and humps and undulatory terrain.

Shelf Ice Sheet

The signature and tonal characteristics are nearly similar to those of polar ice because the surface is covered with the drift ice particles. The shelf ice sheet is clearly decipherable from continental polar ice sheet by its bland featureless, even tone. Only features contained on the surface are the summer melt water channels and cracks produced by the tidal oscillations and pressure subjected by the surrounding polar ice sheet (Coffey, 1975).

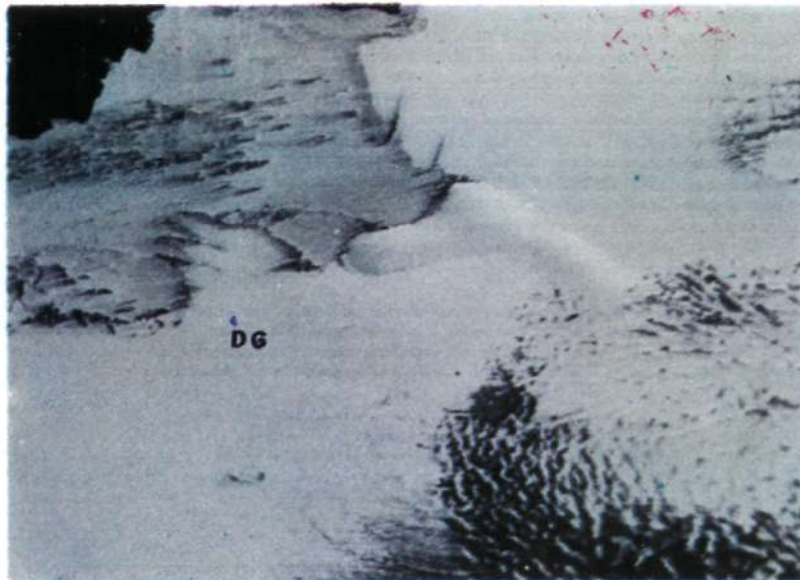


Fig: 5. Ice shelf region around Dakshin Gangotri (DG). The polar ice is in dark and possesses numerous fractures

Blue Solid Crystalline Ice Fields

These ice fields recorded relatively high scatter in band four i.e. 0.5 to 0.6 micro metres wavelength band. The absorption in band 5 i.e. 0.6 to 0.7 micro meters is pronounced. But reflectivities still are very low in bands six and seven (Fennig et al., 1982). In the imagery, these crystalline ice fields appear in darker tones in band 5 and darkest in band 7. The light in band 4 wave lengths scatters along the crystal lattice boundaries and registers high radiance levels, and the red and visible infra-red wave lengths are absorbed. The pure ice crystals consisting of water molecules absorb these wavelengths. The relatively high scatter caused over and above the water signature level is due to the scatter occurring between the crystal boundaries (O'Brien and Muris, 1975). This property clearly aids in demarcating the crystalline blue ice fields which represent negative ice budget regimes, where the area is continuously scoured by winds (Fig. 6).

Crevasses, Glacial Flows and Morainal Fields

Using the four band information, the crevasses, fracture zones in ice sheet, glacial flows and morainal deposits and their trails have been demarcated. The alignment of the erosional features provides good guidance to mark the fractures and large scale crevasses. The glacial flows are typically marked by the alternating bands of exposed crystalline ice and drift ice. These flow bands are very good indicators of the glacier flow and direction, which chiefly cut through the mountain range. The very dark tones occurring within these bands are the morainal boulders deposited by the glacier (Fig. 7).

Exposed Landmass

The exposition of the ice free landmass area in a given image is very small. The low reflectivity area contrasts with the very highly reflective snow fields. The rock exposures and mountains are clearly recognisable in all four bands, from their medium tone on sunlit side and strong shadows cast on the opposite side. Computer aided classification has been separately carried out for lithological units delineation.

Multispectral Digital Analysis of Landsat MSS Data

The ice field values, in general, ranged from 160 to 250 digital counts (brightness values). The exposed mountain range values are between 30 and 100 counts. Sector-wise digital analysis has been conducted isolating the general ice sheet and the mountain regions for geological studies. Band ratioing has been carried for Bands 7 and 5 and the resulting data has been contrast stretched and the false colour images produced. The features which were subdued in the original images and sometimes ambiguous, have come to the fore, enhancing the capability to identify and delineate the following:

1. The small ice plates locked in the fast ice emerged with extreme clarity, where the relative thickness is perceivable. On the leeward side, the non deposition



Fig. 6. Blue Ice field North of Gruber massif. The Pink indicates the drift ice cover on the windward side of the massif

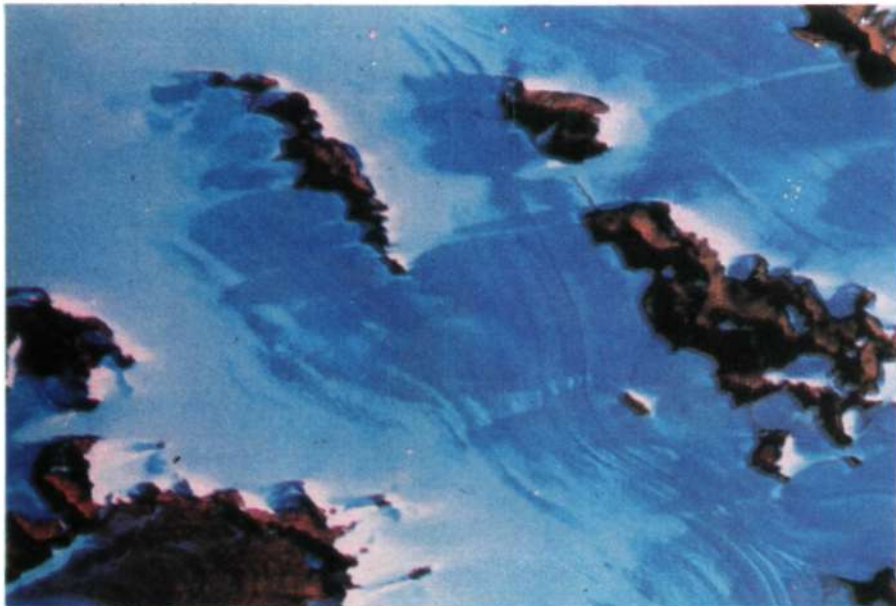


Fig. 7. Glacial flow-marks in Humboldt region. The white streaks indicate the drift ice filled crevasses.

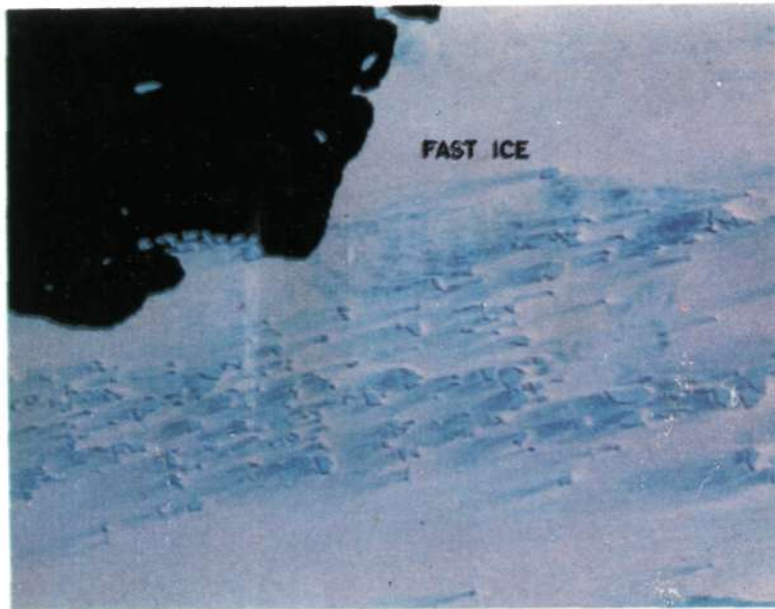


Fig 8. Fast ice with old ice blocks locked up in it.



Fig. 9. Schirmacher bills region depicting blue ice ,flow> lines, Nunataks and driftsnow.

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Fig. 10. Basement ridge extensions of Petermann ranges and huge crevasses, where greater thickness of ice occurs.

area which is ablated has been brought out with great detail (Fig. 8).

2. In the Schimacher region, the Nunataks, the crystalline ice field, the continental glacier flow marks and fractures in the crystalline ice field have been brought out (Fig.9).
3. In some parts of the Wohlthat mountains, the underlying topography is discernible as indicated by the accumulation of drift snow, density and the reflectance increase, indicating the elongated ridges or isolated patches (Fig. 10).
4. The drift snow accumulation on the windward side of the mountain chain contrasts with the surrounding crystalline ice field (Fig. 11).



Fig. 11. The drift ice accumulation: in pink; The transverse fractures are in the Blue Ice Field; The mountainous region: in dark red

5. The continental glacier movement is very clearly demarcated and the morainial deposits marked along with flow marks.
6. The crevasses and fractures were brought out with better clarity. The regions marked free of crevasses, showed up numerous fractures on the ratioed and enhanced image.

On NRSA Multispectral Data Analysis System, digital analysis, extraction of the spectral signatures corresponding to the areas where ground truth and signatures were collected through aerial surveys, have been carried out. The ground data available for various rock types has been compared with the extracted spectral signatures and the classification has been carried out. The signature combinations have been used to carry out the geological classification of the rock types in the eastern Wohlthat mountain region. The geology of Schirmacher hills is well mapped and the three major rock suits have been classified broadly, which match with the geological map.

Schirmacher Hills

The bed rock exposure in Antarctica is expressed in the form of mountain chains, hills and nunataks. These stand out in the form of islands surrounded by ice fields, where the ice is not sufficiently thick and the glaciers have not covered the bed rock or where the glacial retreat has occurred. Schirmacher hills is a small range of low lying hills with about 220 metres high peak and is situated between the shelf-ice and the continental ice sheet. It forms a small obstruction, a basement high, to the flow of the glacier with a length of about 16 km and a maximum width of 3.7 km.

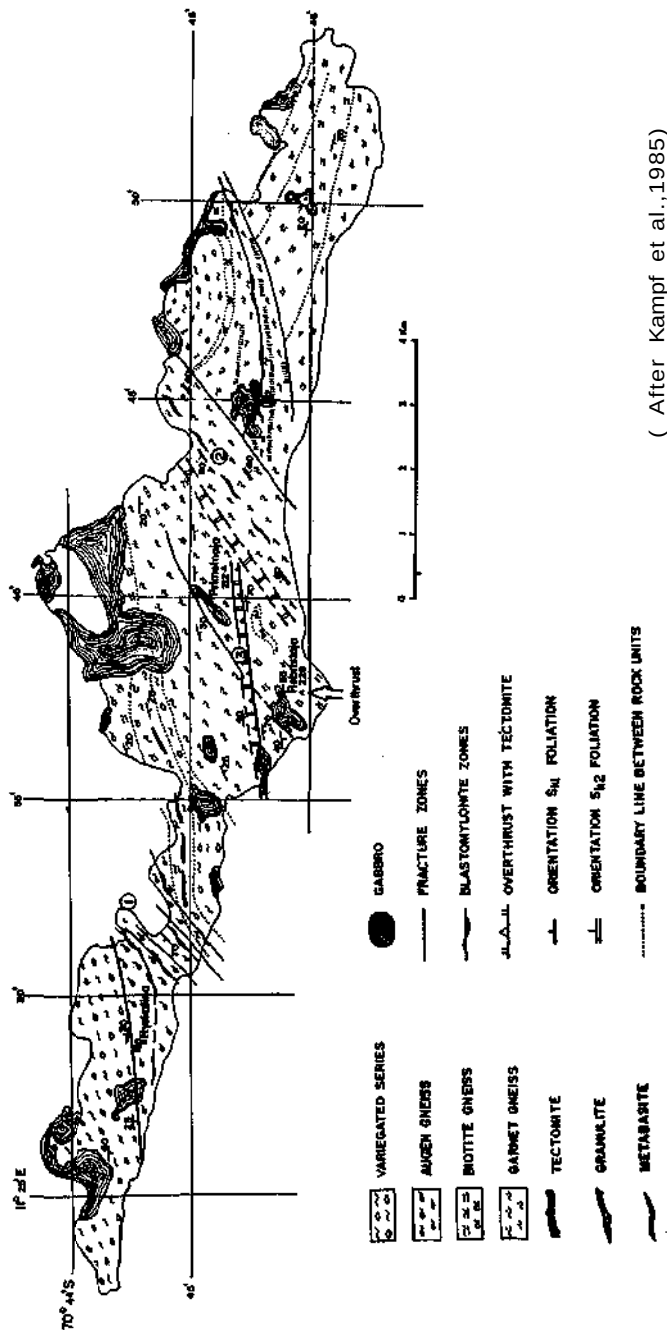
The continental glacier which has retreated, overhangs on the southern side. On eastern and western margins, the glacier flows northward and thrusts against the shelf-ice. Generally this area forms a part of East Antarctic platform. It comprises mainly of medium grade metamorphics, polymetamorphic sequences and relics of high grade metamorphics. The chief lithological units are banded gneisses, augen gneisses, garnet-biotite gneisses, migmatites, quartzofeldspathic gneisses, amphibolites and dolerite dykes. The setting of Schirmacher hills influences the glacial morphology in this region. The northern edge of it lies at the shelf-ice boundary. It presents the wave cut features with bays and lagoons. The northern face is vertical and steep. The shelf-ice is piled up due to thrusting of the polar ice from both east and west directions. The morainal deposits are intercalated and on the southern side the continental glacier overhangs.

The geomagnetic, structural and geological maps have been presented by Bormann *et al.* (1986), geological map by Kampf and Stackebrandt (1985) and also by II/III Indian Antarctic Expedition teams (1985 and 1986). The geological map from Kampf and Stackebrandt has been reproduced (Fig. 12). The classified and pseudocoloured digital output, (Fig. 13), is presented with the classification (Fig. 14). The linear features could not be picked due to their small scale spatial distribution. Only two major fault zones and wide blastomylonite zone could be delineated.

Thermal Mapping of Schirmacher Hills

The Precision Radiation Thermometer was flown across Schirmacher hills along 14 profiles. Fig. 15 presents the isotherm distribution of Schirmacher hills.

GEOLOGICAL SKETCH.MAP OF THE SCHIRMACHER OASIS, DRONNING MAUD LAND



(After Kampf et al.,1985)

Fig. 12

The leucocratic, microcline/plagioclase banded gneisses recorded temperatures between 2 and 5 degrees C. Variations of the biotite gneisses ranged from 6 to 9 degrees C. Garnetiferous banded gneisses and amphibolites ranged from 10 degrees to 13 degrees C. The metabasites ranged between 14° and 18°G. Only isolated patches of the metabasites were recorded above 18° and 20°C in western part of Schirmacher hills.

The isotherms trend mostly in NE-SW direction conforming to the strike directions of the rock sequences and fault directions. In profiles 7 and 8, the 15° and 17°C contours represent the dolerite dykes cutting across the central portion of Schirmacher hills. The variation of temperature from 15 to 20 degrees C in profiles 2, 3 and 4 correspond to the metabasic intrusives which are controlled by the fault tectonics.

Ice Thickness

The profile E-F (Fig. 16) crosses the hump demarcated on Landsat imagery (Fig. 1) and the zone of heavy crevassing occurring along the track of the profile. The section between Schirmacher hills and Gruber rhassif presents numerous crevasses and ogives. Transverse fracture systems have been noted and photographed while traversing from Schirmacher hills to Gruber massif. On the Landsat imagery the curvilinear crevasse patterns are due to the thrusting of the glacier at the eastern part of Gruber massif flowing in a NW direction. The continental glacier south of Schirmacher hills meets this glacier indicating a major depression in the region. It is thus inferred that the heavily crevassed regions are indicative of deep seated basements. This is supplemented by the observation that immediately south of Schirmacher hills, crevassing is minimal although ice thickness is of the order of 1000 metres.

Morphology of Dakshin Gangotri Environs

The morphology of DAKSHIN GANGOTRI environs presents spectacular flow features originating from the Wohlthat mountains where flow direction is from south to north following the topographic low, two large lakes and numerous small lakes, wind driven ice formations, sastrugis (linear ice deposits akin to sand dunes of the deserts), rock formations, fissured and crevassed shelf-ice region, sea-ice. pack embedded with trapped ice-bergs in it, and a large grounded ice-berg (Figs. 17 and 18).

Fast Ice

The fast ice forms from the freezing of the sea surface. During winter it extends up to 60 degrees South latitude girdling the Antarctic continent. During summer it melts off and its remnants are found attached to the permanent shelf-ice. Fast ice patches (Fig. 17 no. 8) remain in the areas protected by the projecting shelf-ice which prevents winds and provides calm sea state region.



Fig. 13. Digital output as obtained from Schirmacher hills.

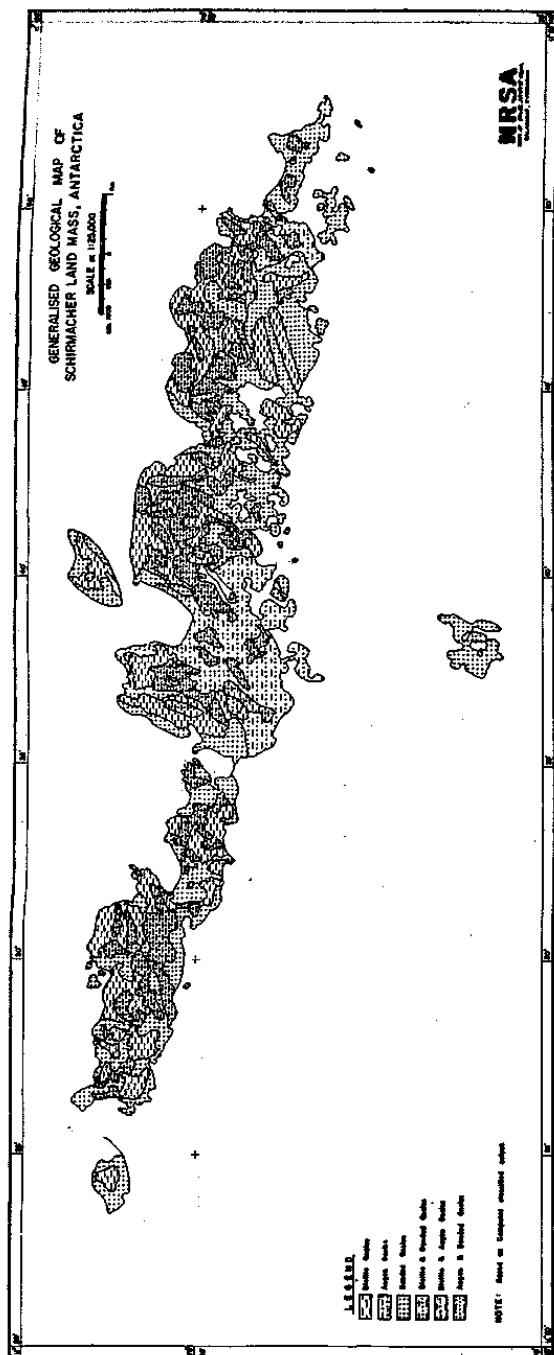


Fig. 14

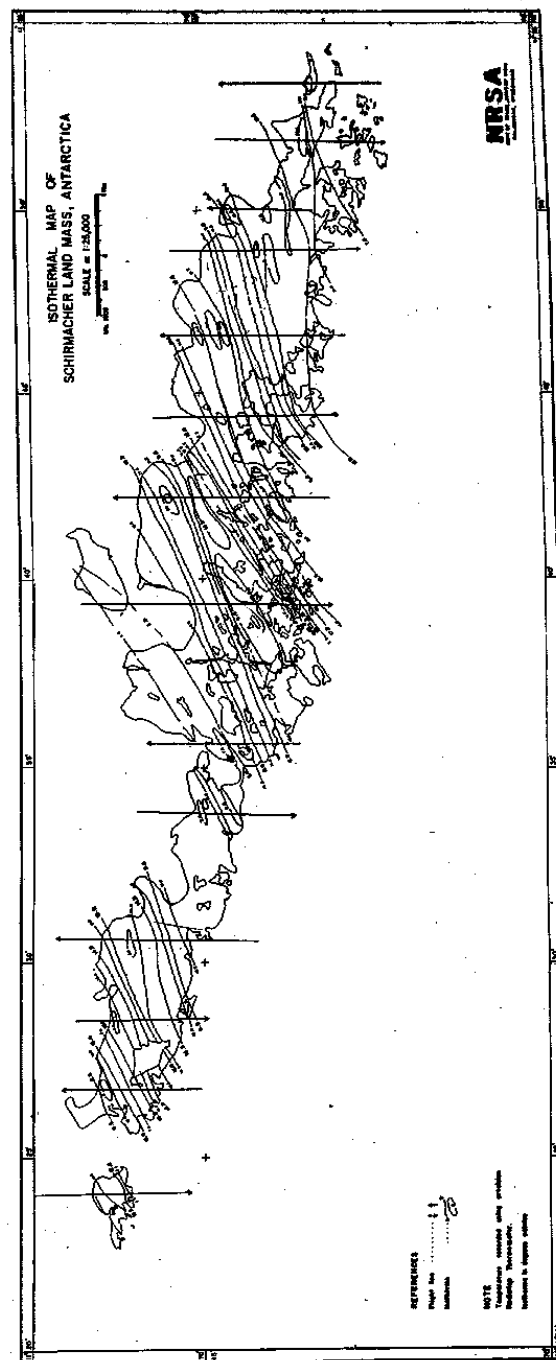


Fig. 15

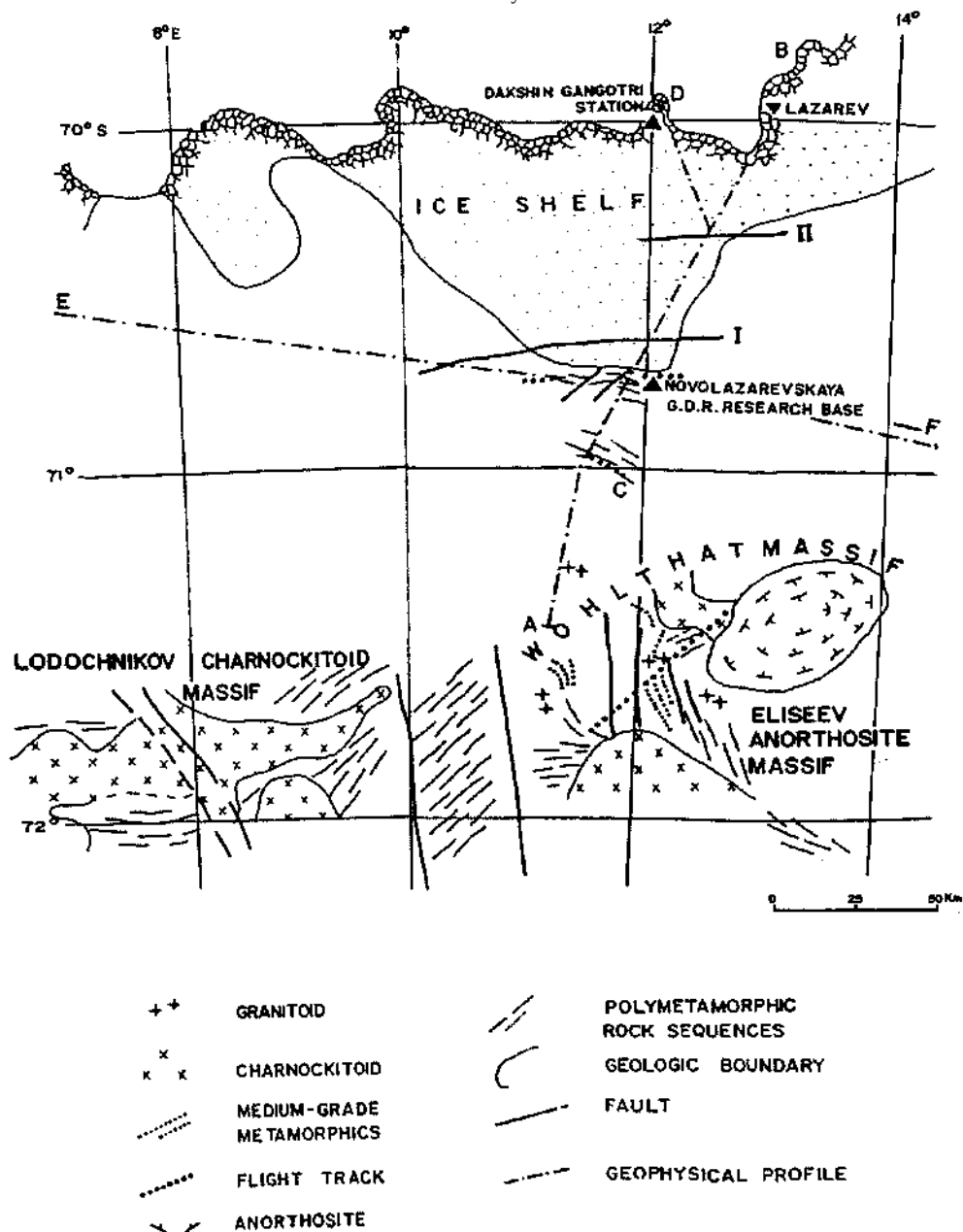


Fig. 16. Generalised geologic map of the central Dronning Maud land (after Bormann et al., 1986)

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Fig. 11. 1. Sea, 2. Ice floes, 3. Fast ice. 4. Over-riding ice plates, 5. Shelf-ice edge. 6. Crevasses along Shelf edge. 7. Shelf-ice region. 8 Permanent fast ice. 9. Grounded Ice. 10. Ridge. 11. Curvilinear flow marks, highly ablated region, with ice dune formations. 12. Gruber massif. 13. Schirmacher hills. 14. Nunataks. 15. Highly crevassed region with glacial flows 16. Blue ice field with drift ice-dunes. 17. Sastrugi field 18 Transverse fractures. 20. Meandering water channels. 21. Continental ice sheet

The old fast ice is interlocked within the sea-ice as seen in Fig. 18. The newly forming fast ice exerts pressure and the old ice blocks are overthrust. On the leeward side, the drift snow does not deposit and winds remove the top cover of drift snow, exposing the year old crystallized ice whose reflectance in band 7 is low and identifiable.

Sea-Ice Pack and Shelf-Ice Cliff Margin

In the NW portion of the imagery is the open ocean water. Absolute dark tone near infrared wavelengths are absorbed within top millimetres of the water

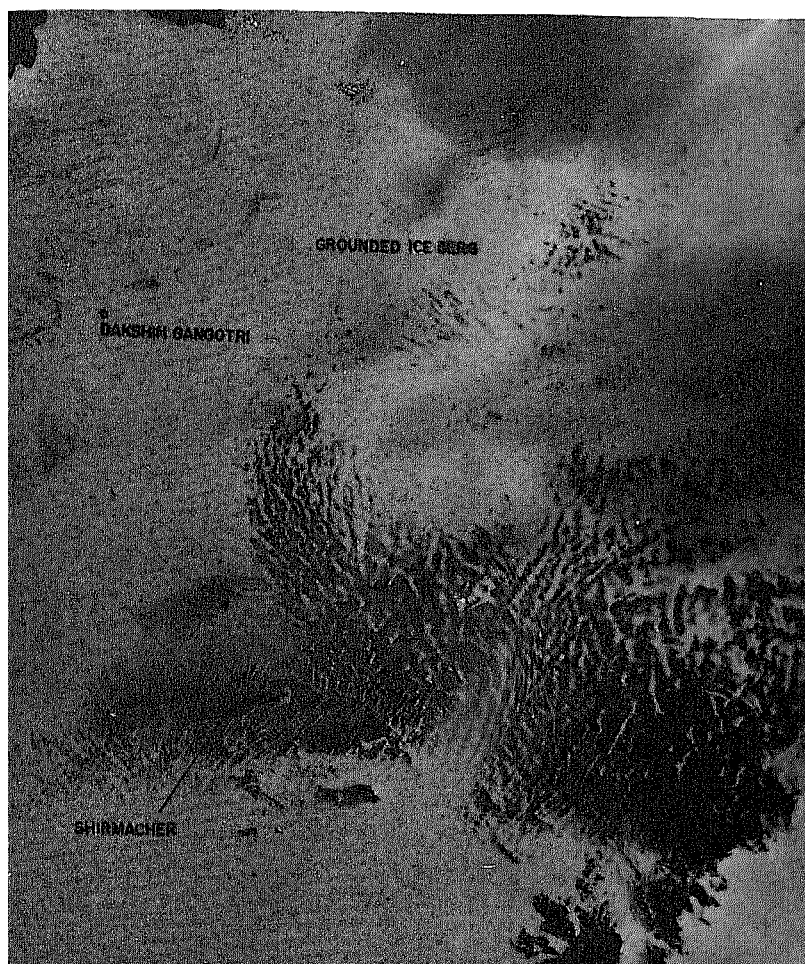


Fig. 18. LANDSAT imagery covering Dakshin Gangotri, Schirmacher hills, and part of Wohlthat Mts. Left corner is oceanic water, followed by the sea-ice with a maximum thickness of 2 m. Locked in it are the old ice blocks. The jagged edge represents the permanent shelf ice edge. The crevasses are developed along the edge. Top right is sastrugi field. South of Schirmacher is the polar continental ice thrusting over the shelf-ice. The flow lines north-of Schirmacher are melt water channels. The curvilinear features indicate the glacier flow directions. The nunataks are south of Schirmacher.

surface. The sea ice has an irregular outline resulting from the breaking and melting of ice.

On surface of this sea-ice pack, long ablation marks on the leeward side of the protruding ice masses are observed. On the leeward side the fresh snow does not occur. The ice-bergs appear to be of various shapes and sizes. From

the ablated region lengths, it is inferred that the protruding ice masses heights are varying. The extreme NW corner of fast ice possesses a clear boundary. This can be attributed to the variation in the fast ice thickness. Near shelf ice cliff a triangular piled up, sloping fast ice patch is clearly visible, adjoining the grounded ice (Leningrad-Kollen).

Wind Generated Drift Patterns

Where the lines are deeply etched and strongly oriented in one direction it can be predicted that heavy winds predominate in that direction. But where the patterns overlies one another it is obvious that one wind direction is not persistent or strong enough to erase the effect of other wind generated flow lines and therefore variability or feableness are predictable (Bagnold, 1954).

Ice-Cliff

The shelf ice abruptly ends into the sea with a sheer vertical cliff with height varying from 20 to 40 metres. Normally, the shelf-ice edge is in balance with the atmosphere and sea action and the advance of the ice-sheet. Calving is one of the chief phenomena by which the shelf-ice is subjected to destruction. Perpendicular, oblique and parallel fissures and cracks are seen all along the shelf. These fissures and cracks are caused by the mechanical tensions in ice. The main causes are 1. tidal action of the sea, which produces tensional fractures and 2. Seaward advancement of the ice-sheet fed by the continental glacier which produces shear fractures.

Though the processes of calving and action of sea waves, spray and winds erode the ice-cliff, continuous movement of the ice mass maintains the equilibrium of the ice-cliff boundary. The coast line as presented in the mosaic from 3 degrees E to 21 degrees E shows two types of ice-cliffs which owe their presence to different processes like, freezing of the ocean which was extending to interior and to the continental ice-sheet which continuously feeds the glacier.

Characteristics of Blue-Ice Field

The highly grooved and scalloped surface, with numerous crack systems and transverse fractures which are typical of blue-ice field occur on leeward side of all the massives in the area. The large extent of blue ice fields in the western part of Wohlthat indicates a negative budget region. The high speed gusty winds carrying very hard drift snow particles scour the area. On the leeward side of mountains and ridges strong ablation acts and deposition of drift ice does not occur. Most of the blue-ice fields are on the leeward side of the mountains. The local precipitation occurs in small quantities. The blue-ice fields also occur on the open continental ice sheet (the area in blue in Fig 1). Alternate exposures of blue-ice occur with the snow drift covered streaks. Transverse fracture systems are located throughout the ice field, one set of which is parallel to the direction of flow of the ice sheet.

The blue solid crystalline ice fields get exposed to the surface with the removal of the top cover by sublimation and ablation leading to glacial retreat. The crystallinity of the ice-field indicates that the fields were once buried. The surface drift snow cover is temporal, being deposited and removed with the gusty winds. The curvilinear pattern observed in Fig. 1 indicates the flow direction of the ice-sheet.

The main continental glacier circumventing the Gruber massif flows in a NW direction. The glaciers cutting through the Petermann ranges which flow along S-N direction are deflected to NW direction. The morainal deposits, in the form of long elongated ridges, are observed in the blue ice-field. The morainal ridges were brought up and exposed along the shear planes of the glacier, which are parallel to the direction of ice-sheet flow (Atenboer, 1964). Both morainal directions and the transverse fractures indicate the ice-sheet/glacier flow directions. The occurrence of moraines have been indicated in Fig. 2. All these moraines are indicative of the directions of flow of the glaciers. The northern (Fig. 1) blue ice-field on continental glacier is crevassed and presents desert features. Snow dunes, barachan type dunes, cirques, wind scoops are present on the ice-sheet.

Ice Features around the Nunataks

The blue crystalline ice fields are exposed on the western side of the nunataks. On the eastern faces of nunataks, as in the case of the high rise mountains of Gruber massif etc., the drift snow piles up. The nunatak rises above the surrounding ice field and is encircled with a deep trench. The bottom is strewn with the angular boulders which have been frost shattered and rolled down. The rock body absorbs the solar radiation, resulting in melting of ice. On the windward side the zone of ice accumulation gradually rises and heaps up, followed by the deep cut trench.

Blue ice-field occurs on the leeward side i.e. on the western side of the nunatak in this region. The extent of the blue ice exposure is related to the size and height of the nunatak. The net result is constant removal of the ice, indicating a negative budget.

Blue ice fields of Wohlthat Mountains

Solid, rock hard crystalline blue-ice fields extensively occur in eastern Wohlthat mountain range. The largest area of occurrence is north and northwest of Gruber massif. This crystalline blue ice field extends half way through to Schirmacher hills and extends upto the shelf-ice and continental ice boundary in the northwest direction from Gruber massif.

The surface is very smooth due to non occurrence of firm ice or drift ice and it is also grooved. Scalloped features are abundant, the circular diameters almost reaching a kilometre as measured on the imagery.

But, in totality, the field is a permanent feature and the surficial changes

occurring are very temporal. The net budget is erosion due to melting, sublimation and ablation. During winters it is net non-deposition and continuous ablation which maintains the character of the solid crystallinity of the blue ice field surface.

Glacial Geology of Wohlthat Mountains

The peaks of Gruber massif and Petermann ranges rise to heights of 2,000 to 2,800 metres. The mountain range widens at Gruber massif ridges of Petermann area and Humboldt massif follow towards WSW. The individual massif almost has a NNW trend and forms a step like feature.

The mountain ranges consisting of Gruber massif and Petermann were surveyed. In fact, Muhlig-Hofmann, Conrad and the Wohlthat mountain ranges act as a dam barring the glacial flow from the plateau region of central Antarctica. Glacial ice-sheet over-rides and cuts through these mountain ranges. The humps are indicators of basement rocks presence but submerged by the glaciers.

The continental ice-sheet cuts and modifies the Wohlthat range to present status as seen in the imagery. It is responsible for all the topographic features of the mountain range apart from the wind, frost and limited chemical weathering agents. The continental ice-sheet flows through and reaches the ocean around in the form of tongues. The general thickness of the ice at the edges is of the order of 200 metres. Where the continental glacier meets the shelf ice, over riding features are quite prominent as noticed near the western end of Schirmacher hills. The continental ice-sheet acquiring gentle slopes merges with the shelf ice and it is difficult to demarcate the transition zone.

The drainage glacier flowing from the plateau east of Gruber massif is mostly un-obstructed except, may be, for the shallow submerged bottom relief of the basement where it is clear that the glacier is over-riding the massif or the mountain range. High amount and frequency of crevassing is noticed on the glacier flowing on the eastern side of Gruber massif indicating deep seated basement. This glacier has, in general, a northerly flow direction, between 16 degrees East and 19 degrees East longitude. The flow of the glacier is trending NNW at the 14 degrees East longitude and the morainal directions are parallel to the flow. A huge morainal deposit north of Gruber massif trends NNW. Numerous humps noted on the Landsat images (Fig. 2) indicate the submerged portions of the bed rock which are continuum of the Wohlthat mountain range. The mountains east of Gruber massif obstruct the flow in the area and cause modifications in the general flow direction (Fig. 2).

From the glacier east of Gruber massif many crevasses and humps have been picked up and the crevasses indicate the occurrence of differential flow in the region. The crevasses trend perpendicular to the flow direction in this area (east of Gruber massif and left of blue crystalline ice field).

The continental ice-sheet gently over-rides the local basement rock. Rises at the eastern part of the hump are lighted by the sun illumination and the

light grey/pink coloured areas demarcate the western slopes on the image. E-W trending bed rock is identified by the light grey/pink colour on the southern side. The top of the hump is quite discernible on the image. Nunataks with greater heights to very low lying nunataks are detectable on the image. Even very small nunataks by their signature could be picked up in the ice sheet, confirmed by comparing the brightness values of all bands for such pixels. They conform to the rock brightness values.

Geology of Eastern Wohlthat Mountains

The Eastern Wohlthat mountains comprise the Gruber massif, three individual massives of Petermann, Humboldt massif and westernmost massif of Orvin Fjella. The distribution and interrelations of different gneisses and lithological units has been attempted through the multispectral digital analysis (Table II). The contact boundaries of the lithounits were brought out prominently. A broad classification

Table II. Mean Brightness values of the Lithounits and Ice types

Type Classified	Band 4	Band 5	Band 6	Band 7	Purity % of Trg. set
Shadow	24.0	18.4	9.7	0.0	100
Lake	184.5	171.0	121.3	43.4	90
	145.9	126.2	80.4	21.3	95
Feldspathic boulders	90.2	116.1	114.7	87.8	87
	100.0	146.1	157.7	123.2	100
Biotite gneiss	65.0	80.1	80.6	60.6	70
	57.4	72.2	73.1	50.3	100
Gneiss	72.2	90.4	94.4	73.3	97
	64.7	48.9	34.3	12.0	87
Anorthosite 1	78.4	63.7	48.9	21.0	70
	66.7	55.2	42.7	18.2	80
Anorthosite 2	59.9	39.9	24.3	4.0	100
Polar ice	242.9	254.0	254.0	196.5	96
	247.8	254.0	254.0	205.5	96
	253.8	254.0	254.0	217.0	97
	254.0	254.0	254.0	252.0	100
Crystalline blue ice	218.1	212.9	161.7	63.0	73
	204.1	201.7	156.7	63.7	75
Blue ice in polar ice field	225.5	240.8	209.0	122.6	83
	205.1	234.2	222.7	153.0	90
Drift snow 1	228.8	234.1	185.5	76.8	60
	221.8	225.3	180.2	75.9	82
Drift snow 2	145.4	157.5	144.4	95.9	100

was attempted by Bormann et al. (1986) for the Gruber massif. The northern half of Gruber massif consists of anorthosite-norite alternations and the lower half, the anorthosite dominated series.

Other rock suites occurring are gneisses, charnockites, gabbros and granitic intrusives. Pegmatite, basalt and dolerite dykes cut across the massives. The computer aided analysis as described in the earlier section was carried out to delineate the lithological units in the eastern Wohlthat mountains. The training sets were fed on the Gruber massif to classify the Petermann ranges and Humboldt massif. The signature clusters were derived and grouped so that the representative lithological units could be classified (Fig. 19).

Green	Biotite Gneiss
Yellow	Gneiss
Pink	Gabbro
Dark red	Anorthosite
Red	Moraines, boulder fields
Ochre	Drift ice piled on windward side of the mountains
Blue	Lakes (Obersee and Untersee)
Black	Shadow

The northwest trending faults are clear on the western flank of the Gruber massif. The gabbros (pink) are in sharp contact with the gneissic intercalations. At the southern margin of Gruber massif, the sharp contact between anorthosite-norites and anorthosites occurs. South of Gruber massif all individual massives appear to be dominated by wholly anorthosite suits. The red indicates strewn boulder fields and morainal deposits.

The rock type of boulders could not be clearly deciphered/grouped because of the intermixing with the drift snow. Hence only a general class to represent the boulder fields/moraines has been made. The northwest and E-W trending basement faults are evident by sharp lithological contacts and the down faulted massifs have been delineated.

In Humboldt mountains the fault zones are predominant and they are even manifested in the character of the blue-ice fields. The moraine and boulder deposits occurring along the western margins of the massives indicate that the boulders were transported and deposited by the glaciers. But frost shattered debris settled at foothills of Humboldt have extensive fields.

FIG. 19 MOSAIC OF A PART OF WOHLTHAT RANGE SHOWING
COMPUTERISED SIGNATURE CLUSTERS



Results and Conclusions

Landsat multispectral scanner data and imagery have been analysed and interpreted to obtain information with respect to morphological features of Antarctic ice-sheet of Queen Maud Land, along Princess Astrid Coast, where Indian Antarctic scientific station Dakshin Gangotri is situated, and for lithological classification of Schirmacher hills and easternmost part of Wohlthat mountain range comprising Gruber, Petermann and Humboldt massives.

The following are the salient features which have been delineated. Solid crystalline blue-ice fields, fracture systems, glacial flow lines, drift ice dunes, crevasses, wind scoops, ice boundaries, topographic highs, drift ice cover over blue ice fields, exposed isolated peaks of hills (Nunataks) or mountains and shallow basement relief. Band ratioing has been carried out for Band 7 and 5 and the resulting data have been contrast stretched and false colour images were produced. These ratioed data have been contrast stretched separately for the ice sheet signature clusters and for the rock exposures.

The features which were subdued in the original images and sometimes ambiguous, have come to the fore enhancing the capability to identify and delineate the following:

The small ice plates locked in the fast ice emerged with extreme clarity, where the relative thickness is perceivable and on the leeward side, the non deposition area and which is ablated has been brought out with great detail. In the Schirmacher region, the nunataks, the crystalline ice field, the continental glacier flow marks and fractures in the crystalline ice field have been brought out. At the Gruber massif, the underlying topography is discernible, as indicated by the accumulation of drift snow density and reflectance increase indicating the elongated ridges or isolated patches.

The reflective properties of snow surface are complicated, depending upon the state of surface, the size of snow grains and the angle of solar incidence and spectral composition of the radiation.

A detailed glacio-morphological map on 1:250,000 scale has been prepared by interpreting the Landsat imagery covering the area between 3 degrees East and 21 degrees East longitude and shelf ice edge in north. Smooth surface manifested by the uniform tone and texture on Landsat imagery south of Wohlthat and Muhlig chain is suggestive of shallow basement occurrence. In particular south of Petermann the basement ridge at near surface is clearly indicated.

The crystalline blue ice fields are exposed at Gruber massif region between Schirmacher hills and Gruber massif, all along Wohlthat mountain range and around nunataks.

The geological classification of Schirmacher hills, and eastern Wohlthat mountains have been attempted. Noritic associations in northern half of Gruber massif and anorthosites in the south have been brought out. The gneisses, charnockites and moraine fields have been classified.

Futuristic Remote Sensing Experiments, Scope and Extent

Utilisation of the Existing Satellite Data

In the present study only Landsat MSS data with four spectral bands and 80 metre resolution has been utilised. It is suggested to use Thematic Mapper data with its 7 band information including thermal band and with 30 metre resolution coupled with SPOT stereo data of 10 metre resolution to provide further detailed information.

Multi Sensor Aerial Surveys

1. Multispectral Scanner Inclusive of Thermal Channel

Daedalus-1268 multispectral scanner is being procured by NRSA shortly. Aerial surveys with MSS and multiband camera can be undertaken for detailed geological surveys. High spatial and spectral resolution would be suitable for detailed geological and structural studies.

2. Side Looking Airborne Radar

NRSA is fabricating the SLAR and it is being readied for the test flight during this year. The SLAR flights could be undertaken at Antarctica for geological and anisotropy studies of the ice.

Radio Echo Sounding for Basement Structure in Antarctica

Geophysical surveys are under way to map the ice thickness, basement depths, structure and fault configuration, to build the tectonic picture of Antarctica. With the Radio Echo Sounding equipment operating between 30 and 300 MHz, the ice thickness measurements can be directly made and thereby map the region accurately for basement depths, to demarcate faulted blocks, ridges, mountain chains covered by the thick ice. Also to determine shelf ice thickness, extent and sub-ice lakes, these studies would provide a complete picture of the basement tectonics. It aids in investigations of bottom roughness characteristics of shelf-ice. Investigations related to the distribution of shelf ice bottom melting and freezing are important to understand and estimate net gain or loss of ice from ice-shelf.

Micro-wave Remote Sensing

1. Altimeter Data

Altimeter data could be utilised to map the entire region of Indian activity. The altimeter data could be processed and superimposed either on TM or SPOT imagery along with other classified information. The altimeter data accurate to 0.25 metres would provide a record of the height information to build a complete picture of the ice topography, ice rises and basins and glacio-morphic units. Future satellite (ERS-I, MOS-II, Radarsat etc.) with altimeter sensors would be of use for Antarctic studies.

2. Synthetic Aperture Radar

The data would be useful in delineating the sea-ice conditions, types of ice etc. The backscatter properties would be useful in determining the anisotropy of the ice by which stress patterns could be delineated. Crevasse and boundaries could be mapped. Roughness of the surface indicative of the erosional activity and thickness of drift snow cover could be determined.

3. Scatterometer

The data could be used to derive the wind speeds for atmospheric studies.

Establishing a Satellite Data Reception Station

It is essential to have our own tracking system to obtain data over part of Antarctica. We are experiencing difficulties in obtaining the TM/Spot data of recent origin covering Antarctica from the distribution centres of EOSAT and SPOT image. The IRS and data reception have been operationalised. Now we have our own resources to obtain remotely sensed data for Antarctic studies and it is essential to establish our tracking system to obtain data over this part of Antarctica and Polar Circle.

The Station could be operated during austral summer and Antarctic coverage could be obtained for further studies. Receiving weather satellite data would give us a means to study this part of Antarctic Circle and South Indian Ocean regime for clouds, sea-ice, sea surface temperatures and currents etc. Its importance with reference to Indian weather is well known and the relation could be studied.

In view of this, it is suggested that the tracking and data receiving station be established at Schirmacher hills where already a permanent base station is being constructed.

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References

- ATENBOER T. VAN (1964): The geomorphology and glacial geology of the Sor-Rondane, Dronning Maud Land, *Antarctic Geology*, SCAR Proceedings, Editor, R.J. Adie.
- BAGNOLD R.A. (1954): The physics of blown sand and desert dunes. Methuen, London.
- BORMANN, P., BANKWITZ, P. BANKWITZ, E., DAMM, V., HURTIG, E., KAMPF, H., WENNING, M., PAECH, H.J., SCHAFER, U. AND STACKEBRANDT, W. (1986): Structure and development of the passive continental margin across the Princess Astrid Coast, East Antarctica, *Jour of Geodynamics*, vol. 6, 347-373.
- FERRIGNO, J.G., WILLIAMS, R.S., AND KENT, T.M. (1982): Evaluation of Landsat 3 RBV images for earth science studies in Antarctica. Antarctica earth science. Ed. Oliver, R.L., Cambridge Univ. Press.
- KAMPF, H. AND STACKEBRANDT, W. (1985): Crustal evolution of the Eastern Antarctic Craton. *Gerlands Beitr. Geophys.*, 94, 251-258.
- O' BRIEN, H.W. AND MURIS, R.H. (1975): Red and near IR spectral reflectance of snow in operational applications of satellite snow cover observations. Ed. A. Rango, NASA Report No. S.P. 391.